

Operators in Tuple

1. Arithmetic:-

```
t1 = (1,2,33)
t2 = (1,2,3)
print(t1 + t2)
print(t1 * 3)

# Output
(1, 2, 33, 1, 2, 3)
(1, 2, 33, 1, 2, 33, 1, 2, 33)
```

2. Assignment:-

```
t1 = (1,2,33)
t2 = (1,2,3)
t1 = t1 + t2
print(t1)
t1 = t1 * 2
print(t1)

# Output
(1, 2, 33, 1, 2, 3)
(1, 2, 33, 1, 2, 3, 1, 2, 33, 1, 2, 3)
```

3. Comparison:-

```
t1 = (1,2,33)
t2 = (1,2,3)
print(t1 == t2)
print(t1 != t2)
print(t1 < t2)
print(t1 <= t2)
print(t1 > t2)

# Output
False
True
False
False
```

4. Identity:-

```
t1 = (1,2,3,4)
t2 = (1,2,3,4)
```

```
print(t1 is t2)
print(t1 is not t2)
# Output
False
True
```

5. Membership:-

```
t1 = (1,2,3,4)
print(2 in t1)
print(5 not in t1)
# Output
True
True
```

6. Logical:-

```
# Non-empty tuple → True
# Empty tuple () → False
t1 = (1,2,3,4)
t2 = (1,2,3,5)
t3 = ()
# and
print(t3 and t2)
print(t1 and t3)
print(t1 and t2)
# or
print(t3 or t2)
print(t1 or t3)
print(t1 or t2)
# not
print(not t1)
print(not t3)
# Output (example)
()
()
(1, 2, 3, 5)
(1, 2, 3, 5)
(1, 2, 3, 4
```